



JAMAICA CONSTABULARY FORCE

Policy and Standard Operating Procedures on the Management of Unmanned Aerial Systems (UAS)

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1.0 Introduction

- 1.1 The dynamic nature and scope of crime and criminal organization has challenged the Jamaica Constabulary Force (JCF) to develop and put into greater use a multiplicity of strategies aimed at increasing its preventative and responsive capabilities. At the same time, technological evolution has positioned the Force to leverage a mix of cutting edge technologies and information management systems in order to manage the changing criminal landscape.
- 1.2 One such effort is to incorporate the use of Unmanned Aerial System (UAS) in the JCF crime management architecture. The adoption and effective use of UAS allow for increased spatial awareness, thus increasing civilian and officer safety when utilized in surveillance, on-going/evolving incidents, search and rescue and other operations where law enforcement’s ‘birds eye view’ is paramount.

2.0 Purpose

- 2.1 This document outlines the JCF’s strategy for the use of Unmanned Aerial Systems. More specifically, it:
 - 2.1.1 establishes the JCF Unmanned Aerial System Unit (UASU);
 - 2.1.2 outlines the JCF’s policy on Unmanned Aerial System; and
 - 2.1.3 outlines the standard operating procedures for the management and use of Unmanned Aerial System in the JCF.

3.0 Scope

- 3.1 This policy and standard operating procedures apply to all sworn and unsworn members’ of the JCF and its auxiliary.

4.0 Abbreviations

CCU	Constabulary Communications Network
GOJ	Government of Jamaica
ICTD	Information and Communications Technology Division
IPROB	Inspectorate of Professional Standards and Oversight Bureau
JCAA	Jamaica Civil Aviation Authority
JCF	Jamaica Constabulary Force
JDF	Jamaica Defence Force
KML	Keyhole Markup Language
KMZ	Keyhole Markup Language Zipped
MNS	Ministry of National Security
NOTAM	Notice to Airmen
PRDB	Planning, Research and Development Branch
SAR	Search and Rescue
SMA	Spectrum Management Authority
UAS	Unmanned Aerial System
UASU	Unmanned Aerial System Unit
UAV	Unmanned Aerial Vehicle

5.0 Definitions

For the purpose of this policy and standard operating procedure the following definitions shall be adopted:

- 5.1 **NOTAM** – Notice to Airmen. A notice released by an aviation authority to alert pilots of potential hazards on a flight route, or at a location that could affect the safety of the flight.
- 5.2 **Pilot** – A member who is trained, certified and responsible for the deployment of UAS
- 5.3 **Spotter** – An assistant to the pilot who is responsible for monitoring the operational area to

ensure that there are no hazards that may endanger the flight, or people who are not part of the UAS flight operation team.

5.4 UAS – Unmanned Aerial System – Comprises: 1) Unmanned Aerial Vehicles (drones), 2) autonomous or human-operated control system, and 3) command and control system.

5.5 UAV – Unmanned Aerial Vehicle – also called drone is a vessel which is flown without a pilot-in-command on-board, and is either remotely and fully controlled from another place (ground, another aircraft, space) or programmed and fully autonomous.

5.6 Payload – The load carried by a vehicle exclusive of what is necessary for its operation

6.0 Responsibilities

6.1 The Deputy Commissioner, Operations Portfolio

The Deputy Commissioner i/c Strategic Operations Portfolio will have overall management of the JCF Unmanned Aerial System Unit (UASU) and shall hold the various heads of the Area Unmanned Aerial Systems Units (UASU) accountable for the use and management of all UAS in their respective areas.

6.2 Assistant Commissioner of Police, Planning, Research and Development Branch (PRDB)

The Assistant Commissioner of Police i/c Planning Research & Development Branch (PRDB) shall be responsible to facilitate research and recommendation of best known practices in the management of UAS in law enforcement.

6.3 The Inspectorate of Professional Standards and Oversight Bureau (IPROB)

The Inspector General IPROB will be responsible for the maintenance of standards in the operation of UAS, and to establish a special panel for investigation of complaint/report management.

7.0 Policy Statements

7.1 The JCF acknowledges that its legitimacy and continued relevance rest heavily on its ability and willingness to innovate and put into greater use new technologies that increase its preventative and responsive capabilities in the management of crime.

7.2 The JCF recognizes that the use of UAS in crime and security management has significant benefits since it offers superior spatial awareness capabilities, minimizing danger to members and civilians while acting as a significant force multiplier.

7.3 The JCF reiterates its commitment to modernization through technology and as such, it is the policy that:

7.3.1 the Unmanned Aerial System Unit (UASU) is hereby established and mandated to manage the operation of all unmanned aerial vehicles (UAVs) and related systems/technologies;

7.3.2 deployment of UAV shall be in compliance with local and international laws (including those concerning human rights), relevant GOJ policies, manufacturer(s) guide, and in accordance with international best known practices;

7.3.3 deployment of UAVs shall be limited to situations in which the benefits derive from its use outweigh the potential risks; these include:

- a. surveillance of high risk areas/targets
- b. specified police operations
- c. surveillance of assets and critical infrastructure
- d. search and rescue (SAR) operations;

7.3.4 UAV will not under normal circumstances be weaponized and/or convey any payload other than that which is necessary for surveillance and/or monitoring;

7.3.5 all communication systems shall be in compliance with the specified spectral band stipulated by the Spectrum Management Authority (SMA) for law enforcement UAV/UAS use; and

7.3.6 only trained and certified members are allowed to operate UAS.

8.0 Standard Operating Procedures

The following are the Standard Operating Procedures of the JCF that members must adhere to in the operation of UAS:

8.1 Acquisition, Maintenance and Disposal of UAS

8.1.1 The JCF will use only approved UAS acquired through the GOJ Procurement Guidelines

8.1.2 Scheduled/preventative maintenance/repair of UAS shall be in accordance with the guidelines, terms or conditions, etc., stipulated by the manufacturer. This is important in order to 1) ensure the optimal functioning and durability of the equipment, 2) minimize the potential for accidents and 3) protect the GOJ's recourse to the manufacturer/supplier in respect of all warranty provisions and entitlements.

8.1.3 Disposal of UAS shall be effected via Asset Disposal Guidelines mandated by the Ministry of Finance with authorization of the Ministry of National Security (MNS)

8.2 Management of Data

8.2.1 Storage/Retention of Data

Retention and handling of recordings from the UAS shall be in keeping with the Evidence (Amendment) Act, 2015, the JCF's Use of Digital Media in Criminal Investigations Policy promulgated in Force Orders No. 3357, Part 1, Sub No. 1 dated October 6, 2011, the JCF Digital Photography SOP, and the JCF's Handling of Exhibits Policy.

8.2.2 Request for Release of Data

8.2.2.1 Request for recordings from UAS must be coordinated through the Constabulary Communications Network (CCU).

8.2.2.2 All access to UAS recordings must be authorized by the Commissioner of Police or his designee.

8.2.2.3 Except where stipulated by law, release of UAS recordings to the public must be after all investigations have been completed, and with due regard to the safety of member(s) involved and the privacy of unconnected persons.

8.2.2.4 Where a decision is taken to release UAS recordings to the public, the footage must be edited in order to conceal the identity of the police, suspects, victims and/or any bystander, as also to preserve the dignity, respect and human rights of persons.

8.2.2.5 Release of UAS recordings must be coordinated through the CCU.

8.2.3 Disposal of Data

8.2.3.1 Where the data has no immediate or potential evidential or law enforcement value or there is no likelihood of court action, the file (footage) may be deleted after one year on the written instruction of the Commissioner of Police or his designee.

8.3 Operational Procedure

8.3.1 Notification to Jamaica Civil Aviation Authority and the Jamaica Defence

Force:

8.3.1.1 A major underlying principle of the operation of UAS is safety. This means safety for the vessel, other users of the airspace as well as those on the ground. The principle underlying operations of UAS in the JCF is that safety should never be compromised. It is the responsibility of the pilot to communicate to the Operation Commander whether or not it is safe to fly.

8.3.1.2 Prior to deployment, once the UAV will exceed 650 feet, the Operation Commander shall direct the Head of the UASU to issue a formal notice to the JCAA and the JDF of their intention to operate UAS in the national airspace. This notification is in keeping with the JCAA's Flight Safety Notification General Guidance in relation to the operation of UAVs (FSN-GN-2015-01 R1 dated April 20, 2016).

8.3.1.3 The notification to the JCAA and the JDF must include:

- i. the area of the operation of the UAS;
- ii. the commencement time of the operation of the UAS; and
- iii. the projected duration of the operation of the UAS

8.3.1.4 Any change in the location, time or duration of the flight must be promptly communicated to the Head of the UASU who will in turn further inform the JCAA and the JDF.

8.3.1.5 Once use of the airspace is concluded, the Operation Commander shall inform the Head of the UASU, who shall formally notify the JCAA and the JDF.

8.3.1.6 Once the notification has been issued, the JCAA will publish the NOTAM via the Aeronautical Information System. The NOTAM informs all parties that an aircraft will be operating in a specific airspace for a specific period of time.

8.3.2 Pre-Trip

Prior to flight, the pilot must gather information about the proposed flight area to ensure the safety of the operation. The pilot must possess the Keyhole Markup Language/Keyhole Markup language Zipped (kml/kmz) file of the location where the aerial operation is to be conducted. Creating a kml/kmz file is critical for submitting the flight advisory to the JCAA and for accountability and tracking purposes.

8.3.3 Pre-Trip Inspection

The pilot should then conduct a pre-trip inspection of the UAS. A pre-trip inspection includes checking battery charge level, checking various mounting hardware, and checking the camera/sensors and storage media to ensure it is sufficient for the task. The pre-trip inspection must be conducted in accordance with the JCF UAS Operations Procedure Guidelines.

8.3.4 Pre-Flight Checks

Pre-flight checks shall be conducted immediately before any flight. The pre-flight checklist repeat some of the pre-trip checklist such as inspecting the UAV and some of the control surfaces. Pre-trip and pre-flight inspections help to ensure airworthiness. This will serve as an early warning sign for both maintenance and mechanical issues, which could substantially affect flight operations. Pre-flight checks must be conducted in accordance with the JCF UAS Operations Procedure Guidelines.

8.3.5 Flight

During flight, the pilot should; 1) continuously scan for airborne traffic or other obstacles, 2) check battery levels constantly and return UAV before it reaches 30 percent of the remaining capacity; and 3) constantly check flight parameters (such as altitude, speed, angle) to ensure that they remain within restrictions and operational goals. Flight checks must be conducted in accordance with the JCF UAS Operations Procedure Guidelines.

8.3.6 Post-Flight

This comprises; 1) the spotter shutting down the UAV, 2) shutting down the controller, 3) inspecting the aircraft, and 4) completing the maintenance, battery and flight logs. Post-flight checklist must be conducted in accordance with the JCF UAS Operations Procedure Guidelines.

8.3.7 Emergency

8.3.7.1 In most emergency situations, the general protocol is to land as soon as is safely possible. This may include landing in areas that were not pre-determined landing zones. The pilot must endeavour to conduct a safe, controlled landing where the unit can be retrieved.

8.3.7.2 During automated flight, if there are concerns that the UAS is not flying the mission planned or that control characteristics are abnormal, the pilot shall take manual control of the UAS and return it to the landing zone if possible. Pilots are required to follow the emergency procedure as outlined in the JCF UAS Operations Procedure Guidelines.

8.4 General Procedures

8.4.1 Training and Certification

The JCF is responsible for maintaining a cadre of trained and certified personnel for the management of the UAS. The JCF will undertake the following:

- a. Restrict use of and access to the UAS to only trained and certified personnel; and
- b. Ensure maintenance of institutional knowledge of the management of UAS.

8.4.2 Security and Storage of UAS

8.4.2.1 During periods when the UAS is not in use, it shall be placed in secured carrying cases and stored at the respective UASU.

8.4.2.2 The designated supervisor for each shift shall be responsible for safe custody of UAS.

8.4.3 Accountability of UAS

8.4.3.1 A master inventory of the Force's stock of UAS will be maintained at the ICTD

8.4.3.2 The UAS Register shall record the issuing and receipt of each UAS

8.4.3.3 The Station Diary shall record the movement of UAS.

8.4.3.4 If a UAS or any part is lost or damaged, member must immediately notify his/her supervisor and document the incident in writing.

8.4.3.5 It shall be the responsibility of the assigned member to ensure that the UAS is operated and maintained according to the manufacturer's instructions and this policy.

- 8.4.3.6 Member shall be held liable for any deliberate, negligent and/or careless handling of UAS which results in lost, damage to unit or property, or injury to personnel.

8.4.4 Damaged, Destroyed, Lost, or Stolen UAS

- 8.4.4.1 In the event of damage, destruction, theft or loss of any UAS, the pilot to whom it was issued shall, immediately or within reasonable time report the circumstances to his/her immediate supervisor.
- 8.4.4.2 The supervisor shall initiate an investigation in the matter and forward a report to the Commanding Officer within 24 hours.
- 8.4.4.3 The Commanding Officer shall then forward a report to the Commissioner of Police and the JCAA within 48 hours.

- 8.4.5** The **UAS Area Registry** shall have the following headings: consecutive number, make and type, serial/model number, unique identifier, number and types of sensors, number and types of associated systems, number of carrying cases, names of trained pilots, number of hours flown, date last serviced, date for next scheduled service, number of batteries and remarks.

- 8.4.6** The **UAS Dispatched Register** shall have the following headings: consecutive number, name and model of UAV, serial number of UAV, number and types of sensors, number of batteries, regulation number and name of person issuing, regulation number and name of person receiving, signature of person receiving, date and time received and remarks.

- 8.4.7** The **NOTAM Register** shall have the following headings: Consecutive number, day, date and time notice issued, name and rank of persons issuing notice, name and position/rank of person in receiving notice and remarks.

- a) All registers shall be official records of the JCF and may be summoned for evidentiary purposes. The registers are to be preserved/maintained in accordance with Force policies.

9.0 Legislative Authorities/Guiding Principles

This Policy and Standard Operating Procedures on Unmanned Aerial Vehicle shall be read in conjunction with the following:

- 9.1** The management of UAS shall incorporate best known practices as advanced by the Jamaica Civil Aviation Authority (JCAA) through the Flight Safety Notification;
- 9.2** General Guidance (FSN-GN-2015-01 R2) for the safe operations of Unmanned Aerial Vehicles (UAV) in Jamaican airspace, revised and published April 14, 2016;
- 9.3** The Civil Aviation Act (2004), the Civil Aviation Regulations (2012), Annex 2 to the Convention on International Civil Aviation in Chicago (1944);
- 9.4** The International Civil Aviation Organization (ICAO) Circular - CIR 328/AN/190 (official publication);
- 9.5** GOJ Policy on Unmanned Aerial Systems (Drone and Drone Support System Policy) Intended for Use in Jamaican Airspace, dated January 1, 2018,
- 9.6** The Ministry of National Security Guidelines for the Unmanned Aerial Systems by the Jamaica Constabulary Force in Jamaica's National Airspace, August, 2017;
- 9.7** The United Nations Department of Peacekeeping Operations' Standard Operating Procedure for Monitoring and Surveillance Technology in Field Missions, established in 2010.

10.0 Relevant Records and Forms

The following records, forms and appendix must be completed or/or updated:

- 10.1** UAS Area Registry

- 10.2 UAS Dispatched Register
- 10.3 NOTAM Register
- 10.4 Organizational Chart UASU – Appendix 1

11.0 Roles and Functions

The following outlines key roles and function:

11.1 Head of the Area UASU shall:

- 11.1.1 take reasonable steps to ensure the security, continuous accountability and proper use of UAS, and manage the UASU in a manner that promotes safety of equipment, safety of other users of the airspace and the general citizenry, and the observation of human rights.
- 11.1.2 have responsibility for:
 - a. maintaining a UAS Area Registry for the accurate inventory of all UAS;
 - b. maintaining a UAS Dispatched Register for the accurate record of ALL movement of UAS;
 - c. maintaining a NOTAM Registry for the accurate record of all notification to the Jamaica Civil Aviation Authority (JCAA) and the Jamaica Defence Force (JDF);
 - d. ensuring that all UAS are serviced and maintained in accordance with manufacturers' guidelines;
 - e. ensuring that UAVs are airworthy;
 - f. authorizing the deployment of UAS;
 - g. ensuring that UAS are used in accordance with guidelines;
 - h. ensuring that only trained UAS pilots are allowed to use the device;
 - i. documenting the use of all UAS under its command;
 - j. implementing measures to ensure accountability for the proper handling, operation, storage and safekeeping of equipment;
 - k. conducting of routine audit and inspection of UAS;
 - l. facilitating external MNS/GOV audits and inspections;
 - m. ensuring compliance with all relevant laws, policies and guidelines and Standard Operating Procedures; and
 - n. ensuring that disciplinary actions are taken against misuse of UAS.

11.2 The **Operation Commander** has overall responsibility for the operational activities within the division. He/she is responsible to:

- 11.2.1 supply timely information to the Head of the UASU on the location, time, duration and any change(s) as it relates to the deployment of the UAS. This information is for onward transmission to the JCAA and the JDF of the JCF's intention to operate UAS in the national airspace;
- 11.2.2 determine what is needed for the operation, brief pilot of flight goals and verify results after landing; and
- 11.2.3 provide feedback to the pilot during flights.

11.3 The Pilot

- 11.3.1 The pilot has the authority on whether or not it is safe for the UAV to take off having taken into account factors such as; weather condition, potential for equipment failure, obstacles/hazards amongst others.
- 11.3.2 If the pilot determines that it is not safe to operate the UAS, he/she shall immediately inform the Operation Commander. The Operation Commander will make the final decision having weighed the risk of operating in the unsafe conditions against the objective of the mission. The pilot is responsible for:

- a. determining if the UAV is airworthy and capable of conducting the proposed operations;
- b. determining if the airspace allows the proposed operation;
- c. determining if the Operation Commander's goals are possible under the conditions presented;
- d. briefing the other members of the flight crew about the mission; and
- e. conducting operations of the UAS within all JCAA guidelines.

11.4 The Spotter

11.4.1 The spotter shall always be a certified pilot and be prepared to take over flight operations if the pilot is not able to complete the mission.

11.4.2 The spotter must remain within direct voice distance of the pilot and has responsibilities for:

- a) monitoring of the operational area to ensure that there are no hazards that may endanger the flight or people who are not a part of the UAS flight operations team;
- b) alerting the pilot immediately if any aircraft comes into the operational area, or if any person or vehicle comes near the operation; and
- c) ensuring that the pilot remains free of interference to safely conduct the flight.

11.5 Information Communication Technology Division (ICTD)

11.5.1 The ICTD shall be responsible for:

- a) recommending the types, specification and capabilities of all UAS to be used in the JCF;
- b) provisioning UAS and ensuring that all UAS island wide are in good working condition;
- c) assigning unique identifier marks on each UAV and related equipment;
- d) disabling stolen, lost and/or any UAS that cannot be accounted for;
- e) submitting annual budget for the acquisition and maintenance of appropriate and adequate UAS;
- f) providing mechanism for the retrieval, storage and admissibility of data as evidence;
- g) ensuring that all communication systems are in compliance with the specified spectral band stipulated by the SMA for law enforcement UAV/UAS use;
- h) recommending the decommissioning and/or disposal of UAS; and
- i) co-ordinating servicing/maintenance/repair of UAS; and establishing and maintaining a UAS Central Register.

11.6 Assistant Commissioner of Police, Planning, Research and Development Branch (PRDB)

The Planning, Research & Development Branch (PRDB) shall be responsible for:

11.6.1 research and recommending best known practices in the management of UAS in law enforcement. Areas for research may include but not limited to:

- a. changes in safety regulations in the aviation industry;
- b. changes in laws, GOJ policies;
- c. new approaches in law enforcement;
- d. cases in the protection of human rights; and
- e. emerging threats in the industry.

11.6.2 maintain data on all reports of misuse, accidents, safety or human rights violations, system failure, loss of equipment and others. These may include but not limited to reports from citizens, the JCAA, the JDF, SMA, other airmen or members of the JCF.

11.6.3 providing official data on reported incidents and to conduct research and recommend changes in strategy, policy or standard operation procedure.

11.7 The Inspectorate of Professional Standard and Oversight Bureau (IPROB)

Inspectorate of Professional Standard and Oversight Bureau shall be responsible for:

11.7.1 Maintenance of Standard

The Inspector General (IPROB) shall have responsibility for the maintenance of standards in the operation of UAS in the JCF.

11.7.2 Complaint/Report Management

IPROB shall investigate and take appropriate actions in respect to all reports of misuse, accidents, near misses, safety violation, human rights violations, system failure, loss of equipment and others. These may include but not limited to complaints from citizens, the JCAA, the JDF, SMA, other airmen or members of the JCF.

11.7.3 Special Panel for Investigation

Where there are reports/complaints such as loss or damage to equipment or property, injury to person, accident or near accident with another vessel, or any serious incident, IPROB shall institute a special panel for investigation. This special panel for investigation may comprise experts from the JCAA, SMA, JDF or a representative of the authorized training entity.

